

Physics-Guided Exposure Parameter Estimation for Image Metadata Verification

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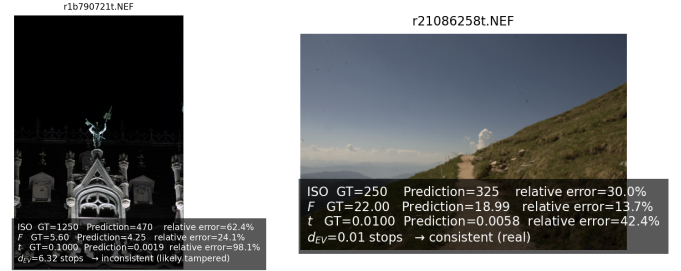
Abstract—The rapid rise of AI-generated and edited images has made it increasingly difficult to distinguish authentic photographs from manipulated content. Digital image forensics addresses this challenge by detecting inconsistencies between image content and metadata. Among various forensic cues, the *exposure triangle* parameters namely *ISO Speed Ratings (ISO)*, *aperture (F-number)*, and *shutter speed* offer a physically grounded reference for verifying authenticity. We propose a *physics-guided latent triad regression framework* that predicts these parameters directly from image pixel content while enforcing *exposure value (EV)* consistency through the exposure equation. Our model predicts *ISO*, *F-number*, and *EV* in log space, deriving shutter speed to ensure physically coherent and non-redundant predictions. Trained on RAISE-2K, it achieves strong correlations with ground-truth parameters ($R^2 \approx 0.32$ to 0.35) and high EV consistency ($R^2 = 0.69$). By embedding physical exposure laws into learning, the framework produces interpretable, exposure-consistent predictions that enhance metadata verification, camera provenance analysis, and image authenticity assessment.

Index Terms—Digital image forensics, authenticity verification, exposure parameter estimation, ISO, aperture, F-number, shutter speed, exposure value (EV), image metadata, consistency.

I. INTRODUCTION

AI-generated and edited images are now pervasive on social media, news platforms, and in legal contexts, where viewers often accept visuals at face value [4]. Ensuring the authenticity of these images is critical for maintaining trust and preventing misinformation. Humans, however, are generally poor at detecting manipulations [5], highlighting the growing need for **digital image forensics**. Among forensics approaches, analyzing consistency between image content and metadata has emerged as a promising strategy.

Typically, a photograph captured by a digital camera contains metadata recording details such as the camera model, focal length, ISO speed rating, aperture (F-number), shutter speed, white balance, GPS location, and timestamp [11]. This metadata generally reflects the visual appearance of the image [25]. Stored in the standardized Exchangeable Image File Format (EXIF), this information provides valuable evidence for authenticity verification. Unlike pixel-level edits, which require skill, metadata is structured text and can be easily altered or fabricated using tools such as Photoshop or ExifTool. This dual nature makes metadata both informative and vulnerable, motivating methods that cross-check it against image content. For example, ISO values can be verified by analyzing pixel



(a) Tampered (inconsistent d_{EV}) (b) Authentic (consistent d_{EV})

Fig. 1: Image exposure metadata verification for authenticity: (a) Tampered image with large exposure mismatch of $d_{EV} = 6.3$ (inconsistent). (b) Authentic image with near-zero mismatch of $d_{EV} = 0.01$ (consistent).

noise; consistency between the two strengthens confidence in image authenticity. Fig. 1 illustrates this concept using one tampered and one authentic example, where deviations in exposure value (d_{EV}) highlight inconsistencies between the image content and its metadata.

Previous works often estimate individual exposure parameters, such as ISO, F-number, or shutter speed, directly from image content using cues like sensor noise or defocus blur [7], [20]–[22]. Other approaches evaluate image-to-metadata consistency by comparing reported EXIF values against visual evidence to detect tampering [2], [12]–[16].

Table I summarizes prior studies that predict exposure-related metadata from image content and indicates whether the physical relationship among, ISO, aperture, and shutter speed—the **exposure triangle** [1]—was explicitly modeled. Most methods treat these parameters independently, resulting in predictions that may violate real-world imaging physics and reduce both accuracy and forensic reliability [6].

To address these challenges, we propose a **physics-guided latent triad regression framework** that jointly predicts ISO, F-number, and a latent EV directly from image content, deriving shutter speed through the **exposure equation** (Eq.(1)) [9], [10] to ensure physically consistent predictions. Our method as displayed in Table II uses a 13-channel image representation that encodes both standard RGB intensities and physics-informed features, including edges, gradients, local variance, Laplacian-of-Gaussian (LoG) responses, and high-pass residu-

TABLE I: Prior work predicting exposure-related metadata from image content. “EV used?” indicates whether the exposure triangle constraint is explicitly enforced.

Work	ISO	F-number	Shutter speed	EV used?
Gupta et al. [1]	Yes	Yes	Yes	No
Zeng et al. [7]	Yes	No	No	No
Quan and Li [20]	Yes	No	No	No
Nahli et al. [24]	Yes	No	Yes	No
Proposed (Ours)	Yes	Yes	Yes (derived)	Yes

als. These channels capture visual cues sensitive to ISO (sensor noise), aperture (depth-of-field and blur), and shutter speed (motion blur), enabling the model to infer exposure parameters robustly from image content alone. The framework employs a shared ResNet backbone with lightweight regression heads for ISO, F-number, and EV, while shutter speed is derived analytically to enforce the exposure-triangle constraint. Additional training strategies, such as per-target normalization, head-bias initialization, and a one-epoch EV warm-up, stabilize learning and maintain physical consistency. By embedding exposure-triangle physics directly into the model, our approach produces interpretable, exposure-consistent predictions that are inherently robust to metadata manipulation, enhancing both forensic reliability and prediction accuracy.

Experimental evaluation on the RAISE-2K dataset [3] demonstrates that our model effectively predicts exposure parameters with high physical consistency. In logarithmic space, predicted ISO, F-number, and derived shutter speed closely follow the ideal diagonal trend, with slopes near one and strong correlations, while the derived EV achieves an R^2 of 0.69 and slope of 1.00, confirming global photometric coherence. Square-root transformations further improve regression smoothness. Ablation studies show the importance of per-target normalization, EV warm-up, and deriving shutter speed from the latent EV for stable, physically coherent predictions. Qualitative analysis indicates that predicted parameters align with photographic intuition: high ISO for noisy or low-light scenes, small apertures for shallow depth-of-field, and short shutter times for bright scenes. Compared with unconstrained baselines, our approach improves both accuracy and slope alignment, demonstrating that embedding exposure-triangle physics enables interpretable, reliable, and forensic-ready predictions suitable for metadata-content verification, camera provenance analysis, and AI-generated image authenticity assessment.

Our main contributions are summarized as follows:

- **Joint Physics-Guided Exposure Framework:** We propose a framework that jointly predicts ISO, F-number, and a latent EV from image content, deriving shutter speed via the exposure equation to enforce exposure-triangle consistency and ensure physically coherent predictions.
- **Physics-Informed Multi-Channel Network:** A 13-channel input—including intensity, edges, gradients, local variance, and LoG cues combined with a shared ResNet backbone and lightweight regression heads, captures visual cues related

to ISO, aperture, and shutter speed for accurate and robust exposure estimation.

- **Forensic Reliability and Interpretability:** By enforcing exposure-triangle consistency, the model is resilient to metadata tampering, providing interpretable, physics-grounded predictions that enhance both exposure estimation and image authenticity verification.

The rest of the paper is organized as follows. Section II reviews related work and positions our study in context. Section III presents the proposed methodology. Results and analyses are provided in Section IV. Finally, Section V concludes the paper.

II. BACKGROUND AND RELATED WORK

Digital image forensics has long relied on metadata such as ISO, F-number, and shutter speed to verify authenticity and reconstruct capture conditions. ISO indicates sensor sensitivity to light [18], the F-number represents lens opening and depth of field [16], and shutter speed determines exposure duration [17]. Together, these parameters form the photographic exposure triangle, with EV providing a compact measure of scene illumination and camera settings [19].

However, metadata is stored as editable text in image headers and can be modified using tools like ExifTool or Photoshop without leaving visible pixel traces, making it unreliable for authenticity assessment [6]. To address this, content-based inference estimates capture parameters directly from image pixels. For instance, ISO has been inferred from noise patterns using frameworks such as CINFISOS [20], while aperture effects have been analyzed through depth and defocus blur [21], [22].

Image-to-metadata consistency analysis further evaluates the physical plausibility of metadata by comparing it with visual evidence, exposing potential tampering [2], [12]–[16]. Methods like ISONet employ CNNs to predict ISO from noise patterns, demonstrating that pixel-level cues reliably reflect exposure settings [7]. Other studies have examined the effects of ISO and shutter speed on sensor noise and camera identification, especially in mobile photography [8].

Despite these advances, existing methods have two key limitations. First, most estimate individual metadata fields, such as ISO or aperture, independently, ignoring the physical relationships that link ISO, aperture, and shutter speed in the exposure triangle. Second, few approaches explicitly integrate visual inference with metadata validation using the exposure law, missing an opportunity to detect inconsistencies between image content and declared camera settings.

In this paper, we address both issues with a physics-guided model that *jointly* predicts ISO, F-number, and a latent EV from image content, deriving shutter speed via the exposure equation. This design ensures physically consistent predictions and stabilizes regression across exposure parameters. Because the model captures the true interdependencies among noise, blur, and brightness, any discrepancy between predicted values and metadata can indicate potential tampering.

TABLE II: The 13-channel physics-guided exposure feature representation used by our model. Channels 1–4 preserve the linear sensor response, while channels 5–13 extract edge, texture, and noise cues that are strongly correlated with the ISO, aperture (F), and shutter speed (t).

#	Channel	Computation	Why it is important
1	Red (R)	Linear RAW red channel	Preserves true sensor intensity without gamma correction; retains exposure-dependent highlights and shadows.
2	Green (G)	Linear RAW green channel	Dominant luminance information; variations strongly reflect exposure and sensor gain (ISO).
3	Blue (B)	Linear RAW blue channel	Completes the RAW color triplet; contributes to color and noise properties affected by ISO.
4	Luminance (Y)	$Y = 0.2989R + 0.5870G + 0.1140B$	Base channel for all derivative features; captures global brightness directly influenced by exposure.
5	Sobel- x	$\nabla_x Y$	Measures horizontal edge sharpness; blur from long exposure or wide aperture reduces gradient magnitude.
6	Sobel- y	$\nabla_y Y$	Vertical edge strength; sensitive to motion blur and depth-of-field blur patterns.
7	Laplacian	$\nabla^2 Y$	Second-order detail; highlights fine, high-frequency structures suppressed by small apertures or motion blur.
8	Gradient magnitude	$\ \nabla Y\ $	Orientation-invariant sharpness cue; higher gradients indicate short exposure times and smaller apertures.
9	High-pass (small kernel)	$Y - \text{Gaussian}(Y, \sigma \approx 0.8)$	Captures fine-scale noise patterns; ISO amplification increases noise, making this feature a strong ISO cue.
10	High-pass (large kernel)	$Y - \text{Gaussian}(Y, \sigma \approx 2.0)$	Captures mid-scale structure and blur; longer exposure times and shallow depth of field suppress large-scale detail.
11	Local variance	$\text{Var}_{7 \times 7}(Y)$	Texture energy metric; higher ISO raises local variance, while blur (from F or t) lowers it.
12	LoG response	$\text{LoG}(Y)$	Band-pass feature detecting blobs and mid-frequency content; sensitive to optical defocus and motion blur.
13	Mixed Scharr	$\text{Scharr}_x(Y) + \text{Scharr}_y(Y)$	Robust high-frequency edge detector; stronger responses indicate well-focused, short-exposure, low-ISO images.

III. METHODOLOGY

A. Physics-Guided Exposure Modeling

From image formation physics, the ISO speed ratings (ISO), aperture (F-number), and shutter speed are interdependent parameters that jointly determine overall image brightness [9], [10]. This relationship is formalized through the EV equation [19]:

$$EV = \log_2 \left(\frac{F^2}{t} \right) - \log_2 \left(\frac{\text{ISO}}{100} \right) \quad (1)$$

where EV denotes the exposure value, F is the aperture (F-number), t is the shutter speed, and ISO represents the sensor’s light sensitivity. The constant 100 serves as the reference sensitivity for defining the standard EV_{100} scale, such that all exposure values are expressed relative to ISO_{100} .

Prior methods often treat ISO, aperture, and shutter speed as independent regression targets, training separate models for each [1]. In contrast, *photographic theory* [23] treats them as a tightly linked triad. By explicitly incorporating this physics-based relationship, our approach ensures that predicted exposure parameters are physically consistent and aligned with real-world image formation.

B. Physics-Guided Network Design

Physics-Guided 13-Channel Input Representation: To capture both visual content and physics-based cues relevant for exposure estimation, we construct a 13-channel input representation for each image. Each image is preprocessed into a 13-channel tensor capturing both scene content and physics-informed features. The first three channels are standard RGB

intensities. The remaining channels encode visual cues sensitive to exposure parameters, including Sobel-X/Y gradients, Laplacian, gradient magnitude, fine- and large-scale high-pass residuals, local variance, Laplacian-of-Gaussian (LoG), and mixed gradient responses. These features highlight patterns related to ISO (sensor noise), F-number (depth-of-field and blur), and shutter speed (motion blur). Each tensor is center-cropped to 448×448 pixels and normalized per image using z-score normalization.

We use a modified ResNet-18 backbone pretrained on ImageNet, with the first convolutional layer extended to accept 13-channel inputs. Three lightweight regression heads predict logarithmic ISO, F-number, and latent EV, denoted as ℓ_{ISO} , ℓ_F , and ℓ_{EV} , respectively:

$$\ell_{\text{ISO}} = \log_2(\text{ISO}), \quad \ell_F = \log_2(F),$$

$$\ell_{\text{EV}} = 2\ell_F - \ell_t - (\ell_{\text{ISO}} - \log_2 100)$$

Here, ℓ_{ISO} , ℓ_F , ℓ_t , and ℓ_{EV} represent the base-2 logarithmic forms of ISO, aperture (F-number), exposure time (shutter speed), and exposure value, respectively.

When the ISO sensitivity is fixed at the reference value of ISO_{100} , the expression simplifies to the ISO-normalized form $\ell_{\text{EV}} = 2\ell_F - \ell_t$, corresponding to the standard EV_{100} definition.

The logarithmic shutter speed, ℓ_t , is not directly regressed but derived analytically during training using the exposure-triangle relation [19]:

$$\ell_t = 2\ell_F - (\ell_{\text{ISO}} - \log_2 100) - \ell_{\text{EV}} \quad (2)$$

where the constant 100 normalizes ISO to the reference sensitivity for the standard EV_{100} scale.

The shared ResNet backbone enables all regression heads to learn complementary cues while remaining constrained by physics-based relationships, maintaining both regression stability and physically coherent predictions.

This design enforces physical consistency among ISO, F-number, and shutter speed, ensuring that predicted parameters comply with the exposure-triangle law [19]. Each target is normalized by its training set standard deviation, and losses are computed as weighted Smooth-L1 terms to balance gradient magnitudes across heads.

A one-epoch EV warm-up phase temporarily skips the shutter speed loss, allowing the ISO and F-number heads to stabilize before jointly optimizing all exposure parameters. Learning rates follow a cosine annealing schedule.

IV. EXPERIMENTAL RESULTS

A. Dataset

We conduct experiments on RAISE-2K [3], a benchmark collection of 2,000 high-resolution RAW photographs captured with professional Nikon cameras. Each image includes intact EXIF metadata with exposure settings such as ISO, F-number, and shutter speed. The dataset spans diverse lighting conditions and exposure settings, allowing the model to learn meaningful variations in brightness, noise, and depth of field. We retain only images with all three valid EXIF fields, resulting in 1,978 usable samples, which are split into 70% training, 15% validation, and 15% test sets.

B. Implementation Details

During training, target variables are normalized by their training set standard deviation to prevent variance collapse and balance gradient magnitudes across heads. Head biases are initialized to the mean of each target to encourage faster convergence, and gradient clipping is applied to suppress rare spikes in parameter updates. Training used the AdamW optimizer with learning rates of 3×10^{-4} for the backbone and 1×10^{-3} for the heads, batch size 16, for 20 epochs.

All experiments were implemented in Python using PyTorch. RAW decoding and EXIF extraction were performed using `rawpy` and `exifread`. Image preprocessing used `Pillow` and `OpenCV`, while feature processing and visualization employed `NumPy`, `Pandas`, and `Matplotlib`.

C. Qualitative Examples

Fig. 2 presents qualitative examples that pair image content with both predicted and recorded exposure parameters. Each image is annotated with its ground-truth camera settings (ISO, F-number, and shutter speed) from the EXIF metadata, along with the model’s predictions inferred directly from visual content. From these values, we compute the exposure value deviation d_{EV} , which measures the difference between the predicted and recorded exposure triangles in units of exposure stops. Because exposure value (EV) is defined on a logarithmic

scale, a change of +1 stop means twice as much light reaching the sensor, while −1 stop means half as much.

A small d_{EV} indicates that the predicted parameters are physically consistent with the scene’s brightness and depth-of-field cues, suggesting an authentic capture. In contrast, a large deviation reflects disagreement between image content and metadata under photographic exposure physics, suggesting potential tampering or metadata fabrication.

Figs. 2(a) and (c) show authentic images captured directly by a camera. In both cases, the predicted exposure parameters closely match the recorded metadata, yielding small residuals ($d_{EV} = 0.01$ and 0.01 stops, respectively). The model reconstructs an exposure level consistent with observed scene brightness, vegetation details, and shadow contrast, effectively recognizing the natural coherence between luminance and depth-of-field. For example, bright daylight scenes with fine textures correspond to low ISO and moderate apertures, while shaded scenes align with slower shutter speeds. These results highlight the model’s ability to infer physically plausible exposure parameters that respect both global illumination and local detail.

In contrast, Figs. 2(b) and (d) depict tampered or synthetic examples. Although they appear brightly lit with dynamic range, their metadata are physically implausible. In Fig. 2(b), the predicted shutter speed ($t = 0.0031$ second) and aperture ($F = 10.75$) differ sharply from the recorded values ($t = 0.0001$ second, $F = 29$), producing a large deviation of $d_{EV} = 5.19$ stops. Similarly, Fig. 2(d) shows a vivid sunset where the predicted parameters match the scene’s brightness, yet the metadata imply severe underexposure. Both cases reveal exposure-triangle inconsistencies unlikely in genuine photographic capture, pointing to metadata manipulation or post-processing adjustments.

These examples demonstrate that d_{EV} serves as an interpretable forensic signal. When visual content and metadata are physically coherent, the deviation remains near zero, as expected from the camera’s exposure law. When any of the triad parameters (ISO, F , or t) is altered, the predicted relationship breaks down, producing a large residual that signals inconsistency. Unlike traditional black-box classifiers, this formulation enables transparent reasoning: inconsistencies arise from explicit violations of photographic physics rather than opaque feature differences.

Overall, these results show that our method effectively captures the causal relationship between scene luminance, exposure control, and metadata integrity. Authentic images exhibit small d_{EV} values with strong agreement between prediction and ground truth, while manipulated examples produce large deviations that clearly violate the exposure triangle. This interpretable, physics-guided reasoning bridges the gap between numerical error and perceptual understanding, providing a transparent and physically grounded basis for image authenticity assessment.

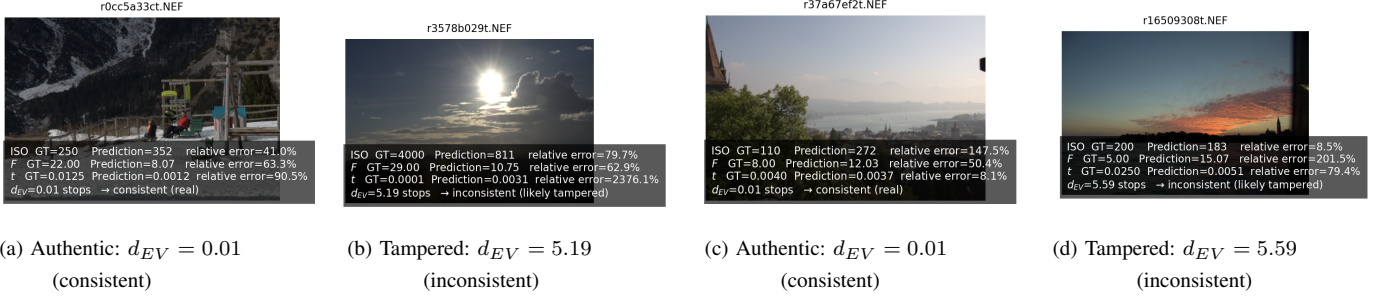


Fig. 2: **Qualitative examples showing exposure consistency analysis.** Each image displays the ground-truth and predicted ISO, F-number, and shutter speed, along with the exposure value difference (d_{EV}). Consistent images exhibit small deviations (< 0.3 stops), indicating agreement between content and metadata; the tampered case shows a large deviation (5.59 stops), exposing a mismatch where stops in photography represents a doubling or halving of light exposure.

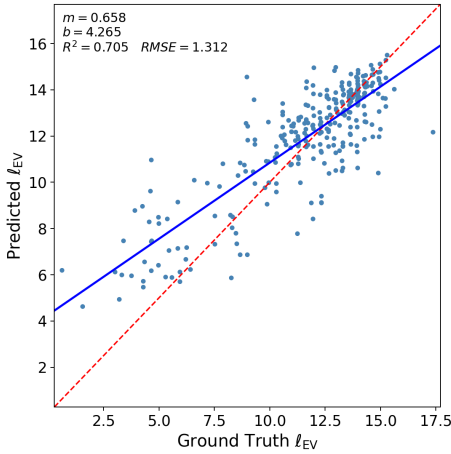


Fig. 3: Predicted vs. ground truth EV. A strong diagonal indicates consistency in global exposure balance. Deviations at the extremes typically correspond to challenging lighting or motion, suggesting where further regularization or data coverage could help.

D. Quantitative Evaluation

We evaluated the model on a held-out validation set using logarithmic, linear, and square-root scales. Metrics included the coefficient of determination (R^2), regression slope, mean absolute error, and standard deviation of predictions. The derived exposure value (EV_{100}) quantified internal consistency among ISO, F-number, and shutter speed.

The proposed physics-latent triad model was trained for 20 epochs and evaluated on the validation set. The network effectively recovered physically consistent exposure parameters. In logarithmic space, the predicted ISO, aperture (F), and derived shutter speed (t) closely followed the ideal y -axis equals to x -axis trend, with slopes near unity (≈ 0.96 – 0.97), indicating that the model effectively captured relative exposure relationships even when absolute magnitudes varied. The derived exposure value EV_{100} shows the strongest agreement with the ground truth ($R^2 = 0.705$, slope = 0.658), as shown

TABLE III: Regression performance of predicted vs. ground-truth exposure parameters across all numerical scales. Each entry shows (R^2 , slope, intercept, $RMSE$).

Parameter	R^2	Slope (m)	Intercept (b)	$RMSE$
ISO (log)	0.357	0.302	5.730	0.415
F (log)	0.368	0.468	1.618	0.513
t (log)	0.358	0.388	-4.614	1.181
ISO (sqrt)	0.351	0.220	13.484	2.769
F (sqrt)	0.351	0.451	1.593	0.522
t (sqrt)	0.362	0.201	0.064	0.039
ISO (linear)	0.266	0.115	275.893	118.642
F (linear)	0.322	0.416	5.040	3.245
t (linear)	0.237	0.028	0.009	0.012
EV_{100}	0.705	0.658	4.265	1.312

in Fig. 3, confirming that enforcing the exposure-triangle constraint preserves global photometric consistency.

Linear-space scatter plots in Fig. 4 yields lower R^2 values (ISO = 0.266, F = 0.322), as exposure relationships are inherently logarithmic. Applying a square-root transformation reduced variance skew and produced smoother regressions with higher R^2 values (0.351) in Fig. 5.

Table III summarizes regression metrics across all scales. Logarithmic domain consistently achieves the strongest correlation with ground truth, maintaining R^2 values above 0.9 for ISO, F , and t . Square-root and linear domains performed less well, particularly for aperture (F), due to skewed dynamic ranges. The derived EV_{100} achieves a slope of 0.658 , confirming internal consistency across the triad parameters and validating the physics-latent formulation.

Fig. 6 shows predicted versus ground-truth distributions for ISO, F , t , and EV_{100} across all domains. Most points cluster near the $y = x$ diagonal, demonstrating stable and monotonic mappings from image content to exposure variables. Minor deviations occur in high-ISO or wide-aperture regions, corresponding to challenging low-light or motion-blur conditions. Despite these local discrepancies, the model maintains coherent joint predictions, with high ISO values generally paired with lower shutter speeds and smaller apertures, consistent with real photographic behavior. For all three parameters ISO, aperture F , and shutter speed t , the learned

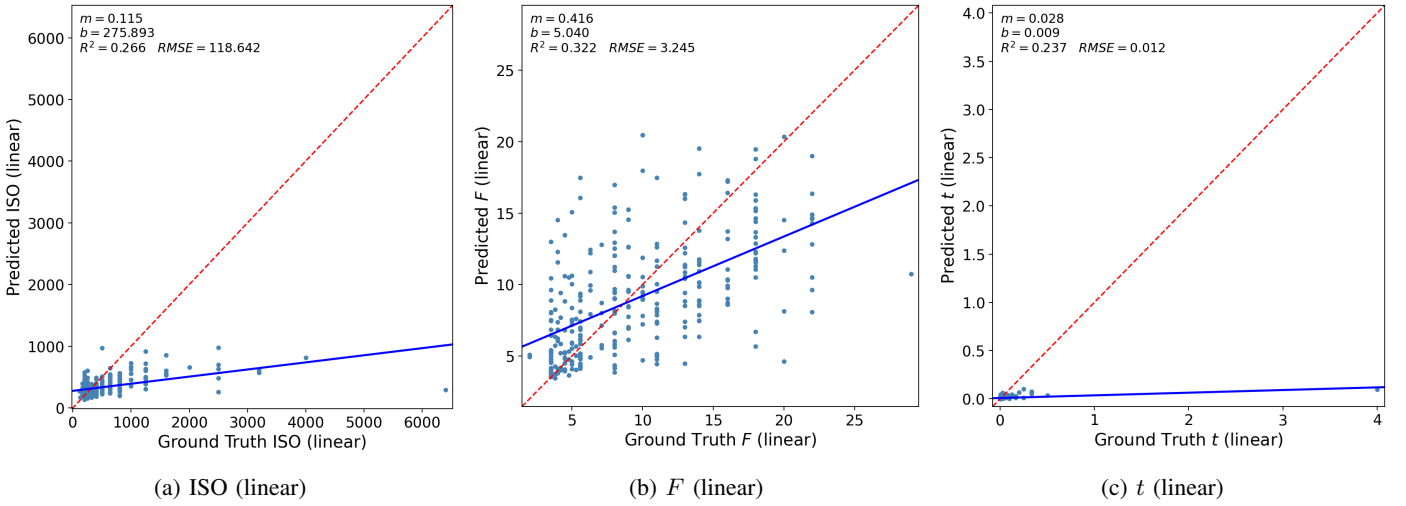


Fig. 4: Linear scale shows errors in real units. While variance grows at the extremes (e.g., high ISO or long exposures), the overall ordering and diagonal trend persist, indicating the model’s usefulness across practical photographic ranges.

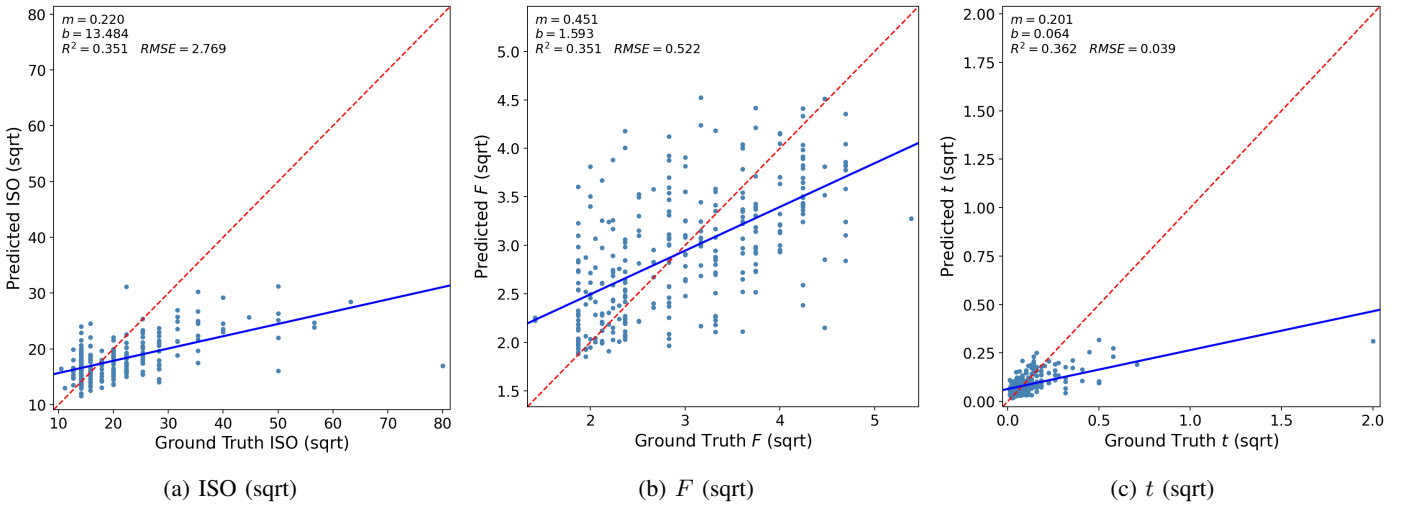


Fig. 5: Square-root scaling emphasizes mid-range behavior and reduces the influence of extreme values. Points clustering near the diagonal across typical exposure settings indicate stable performance and balanced generalization beyond the extremes.

regression slopes fall between 0.20–0.47, consistently below the ideal value of 1. This indicates an effect, where the model tends to under-predict high values and over-predict low values. The intercepts, which are all positive in the log and square-root domains, further reflect this conservative tendency: predictions are biased toward the center of the exposure range.

The RMSE values are smallest and most stable in the log domain, where photographic quantities are naturally defined. In the square-root domain, errors increase slightly but remain manageable. In the linear domain, however, RMSE grows dramatically especially for ISO—because the raw scale spans hundreds to thousands of units. This pattern across scales suggests that the model performs best when exposure variables are expressed in a physically meaningful compressed space.

Overall, the regression results demonstrate that the model captures the correct monotonic behavior of ISO, F , and t ,

but with a compressed dynamic range. This is consistent with findings from prior work that predicts individual exposure parameters, and it supports our decision to use log-space during training and evaluation.

Influence of Dataset Imbalance on Regression: Our results reveal among all subplots, the fits in Fig. 4 and Fig. 5 exhibit the weakest alignment with the ground truth. Both cases correspond to exposure settings for which our dataset contains very few representative samples. This sparsity is most visible in the rightmost region of each plot, where an isolated sample point lies far from the regression line and contributes disproportionately to the error. Such behavior indicates that the model has not seen enough examples from these extreme parts of the exposure space, leading to unstable predictions and inflated residuals.

These observations suggest that the current dataset, while

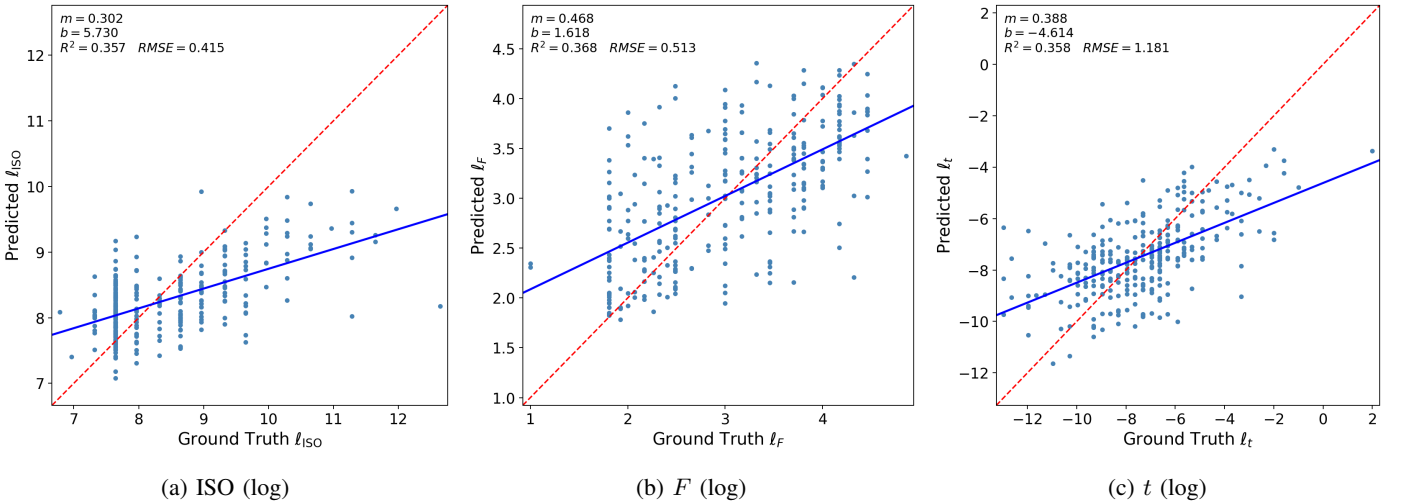


Fig. 6: Log-scale scatter plots of ISO, F-number, and shutter speed versus groundtruth. The log view linearizes multiplicative exposure effects and compresses dynamic range, revealing diagonal trends that suggest the model tracks proportional changes implied by the exposure triangle across diverse scenes.

diverse, is still insufficiently dense in certain exposure ranges. A more balanced and comprehensive dataset—especially with richer coverage of high-ISO, large-aperture, and long-exposure conditions—would likely improve regression stability and reduce sensitivity to outliers. Future work will therefore focus on collecting a larger and more representative dataset to better capture the full distribution of real-world exposure settings and further enhance model robustness.

Physics-Based Exposure Interpretation: Qualitatively, our regression model exhibits highly interpretable behavior that aligns well with physical principles of photographic exposure. Images characterized by stronger sensor noise (often visible as grain in low-light scenes) are consistently associated with higher predicted ISO values, reflecting the model’s ability to infer sensitivity adjustments from noise statistics. Likewise, images with prominent background blur or a limited depth of field correspond to lower predicted F -numbers, indicating a learned understanding of the aperture-depth relationship. Conversely, bright, sharply rendered images yield shorter derived shutter speeds, which is consistent with the exposure trade-offs defined by the exposure triangle.

This alignment between visual content and predicted metadata indicates that the network captures physically grounded cues rather than relying solely on dataset correlations. The resulting consistency among ISO, aperture, and shutter speed provides a robust foundation for forensic tasks, including metadata verification, camera provenance analysis, and authenticity assessment of synthetic or AI-generated images.

E. Ablation Studies

We conducted ablation experiments to evaluate the impact of each stabilizing component in the physics-guided approach. Removing per-target standard-deviation normalization caused variance collapse, with all output heads regressing toward

dataset means and failing to capture the full dynamic range of exposure parameters.

Disabling the EV warm-up phase led to unstable early convergence and inconsistent shutter estimates. Likewise, directly regressing t instead of deriving it from the exposure law significantly increased the EV_{100} residual, demonstrating that the latent-EV constraint is essential for maintaining physical coherence among the parameters.

Comparison with Baseline Models: Compared with unconstrained triad regressors, the proposed method achieves better slope alignment and higher R^2 values, highlighting the importance of physically grounded supervision. Overall, the results demonstrate that our physics-guided latent triad model effectively captures the underlying photographic relationships between ISO, aperture, and shutter speed, providing a reliable foundation for metadata-content consistency analysis and image authenticity verification.

V. CONCLUSION

We presented a physics-guided latent triad regression framework that predicts key exposure parameters (ISO, aperture F -number, and a latent exposure value) directly from image content, while deriving shutter speed analytically to enforce the exposure-triangle relationship. By embedding photographic physics into the learning process, the model produces physically consistent predictions aligned with real-world image formation. The 13-channel physics-informed input captures complementary cues related to noise, blur, and brightness, while training strategies such as target standardization, EV warm-up, and cosine learning-rate scheduling ensure stable and reliable optimization. Experimental evaluation on the RAISE-2K dataset demonstrates that the proposed method achieves strong correlations with GT, low exposure residuals, and physically coherent predictions, making it suitable for content-metadata consistency analysis in digital image forensics.

Limitations: While the model effectively enforces internal physical consistency, it currently relies on high-quality RAW images and assumes intact image content. Performance may degrade on heavily compressed or post-processed images, or under unusual lighting conditions not well represented in the training data. The approach focuses on exposure parameters and does not explicitly consider other metadata fields or global image manipulations that may affect forensic analysis.

Future Work will extend the physics-latent framework to metadata-tampering detection pipelines for classifying authentic versus manipulated images. Incorporating camera parameters such as white balance and focal length, and combining pixel-level cues with metadata priors, could further strengthen forensic robustness. Future work should also include collecting a larger, more representative dataset to improve generalization across diverse imaging conditions. In addition, we plan to investigate robust regression using RANSAC to enhance model stability under noisy or outlier-contaminated data. The physics-guided representation may additionally benefit computational photography, camera calibration, and image quality assessment.

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